JUNMO KOO

Division of Mechanical Engineering, Korea Maritime & Ocean University

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EDUCATION

Korea University

PhD Mechanical Engineering Aug 25, 2020

BS Mechanical Engineering Aug 25, 2014

PhD Advisor: Prof. Joon Hyung Shim

Thesis: Atomic Layer Deposition of Metal Nanoparticles as Fuel Cell Catalysts

CAREER EXPERIENCES

Medilayer

Founder & CEO 2026 – present

Particle ALD-based nanocoating service for powder-type pharmaceuticals

Selected for the 2026 Pre-Startup Package

Selected for the 2026 Busan Startup Accelerating Program

Division of Mechanical Engineering, Korea Maritime & Ocean University

Assistant Professor Mar 1, 2024 – present

Semiconductor Manufacturing Process (ALD, ALE, CVD)

Hydrogen energy (fuel cell, hydrogen production)

Smart Grid (techno-economic analysis, energy source prediction)

Battery (lithium metal, molten-salt thermal, all-solid-state batteries)

Department Chair Feb 1, 2026 – Jan 31, 2028

Department of Refrigeration & Air-Conditioning Engineering, Graduate School

Head Professor Feb 1, 2026 – Jan 31, 2028

Refrigeration & Air-Conditioning Engineering Major, Division of Mechanical Engineering

Materials & Energy Technology Center, Agency for Defense Development

Senior Researcher Apr 1, 2023 – Feb 18, 2024

Thermal battery for missile power source

Fuel cell/battery hybrid energy system for unmanned underwater vehicle

Energy Materials Research Center, Korea Institute of Science and Technology

Postdoctoral Researcher Sep 1, 2022 – Mar 31, 2023

Hydrocarbon-fueled solid oxide fuel cell (SOFC) stack

Logic CVD, Samsung Electronics Semiconductor R&D

Staff Engineer Sep 1, 2021 – May 15, 2022

Hard mask patterning film deposition processes

DRAM Diffusion, SK Hynix R&D Center

TL (CL4) Aug 18, 2020 – Jul 18, 2021

SiN & SiO_2 deposition processes for DRAM cell-bit line

Gap-fill material deposition processes

Renewable Energy System Lab, Korea University

Research Assistant Sep 1, 2014 – Aug 17, 2020

Carbon-coking resistant electrodes for direct methanol solid oxide fuel cell (DMSOFC)

Extremely stable electrodes for polymer electrolyte membrane fuel cell (PEMFC)

Fuel cell performance prediction model for unmanned underwater vehicle

High performance, platinum-free electrodes for anion exchange membrane fuel cell (AEMFC)

Intrinsic mechanical properties of nanoscale thin films by tensile testing on water

surface

In-house thermal/plasma-enhanced ALD systems and processes

Various ALD recipes with in-house ALD system: metal (Pt, Pd, Ag) and metal oxide (ZrO_2, HfO_2, Al_2O_3, SrO, TiO_2, BaO, Y_2O_3, ZnO, SnO_2, La_2O_3, Co_3O_4)

In-house aerosol-assisted chemical vapor deposition (AACVD) system and processes

Thin film coating processes by inkjet printing

HONORS & AWARDS

Outstanding Researcher Award

KEPCO Research Institute Dec 31, 2020

Best of Energy Technology, 2019

Energy Technology Feb 26, 2020

Best Paper Award

Korea University Graduate School Feb 25, 2020

Best Paper Award

Korea University Graduate School Feb 25, 2019

Best Paper Award

Korea University Graduate School Feb 25, 2018

Best Presentation Award

7th East Asia Mechanical & Aerospace Engineering Workshop Nov 28, 2017

ADVISEE HONORS & AWARDS

Excellence Award (2 teams), Capstone Design Competition

Korea Maritime & Ocean University May 2026

Excellence Award, Capstone Design Competition

Korea Maritime & Ocean University May 2025

TEACHING

Fall 2026	Engineering Mathematics II		
Fall 2026	Fuel Cell		
Spr. 2026	Renewable Energy	Enrollment: 44	Evaluation: 4.5/5.0
Spr. 2026	Engineering Mathematics	Enrollment: 43	Evaluation: 4.5/5.0
Spr. 2026	Basic Electric and Electronics	Enrollment: 38	Evaluation: 4.6/5.0
Spr. 2026	Int. to Ref. & A/C Eng. (Team Teaching) Evaluation: N/A		Enrollment: 39
Spr. 2026	Capstone Design II	Enrollment: 12	Evaluation: N/A
Fall 2025	Engineering Mathematics II	Enrollment: 34	Evaluation: 4.4/5.0
Fall 2025	Statics (003)	Enrollment: 24	Evaluation: 4.3/5.0
Fall 2025	Statics (004)	Enrollment: 24	Evaluation: 4.1/5.0
Fall 2025	Capstone Design I	Enrollment: 6	Evaluation: N/A
Spr. 2025	Int. to Ref. & A/C Eng. (Team Teaching) Evaluation: 4.0/5.0		Enrollment: 45
Spr. 2025	Engineering Mathematics	Enrollment: 42	Evaluation: 4.5/5.0
Spr. 2025	Basic Electric and Electronics	Enrollment: 41	Evaluation: 4.6/5.0
Spr. 2025	Renewable Energy	Enrollment: 56	Evaluation: 4.3/5.0
Spr. 2025	Semiconductor Process & Equipment	Enrollment: 41	Evaluation: 4.2/5.0
Spr. 2025	Capstone Design II	Enrollment: 6	Evaluation: N/A
Spr. 2025	Advanced Semiconductor Process*	Enrollment: 3	Evaluation: N/A
Fall 2024	Statics (007)	Enrollment: 28	Evaluation: 4.4/5.0
Fall 2024	Statics (002)	Enrollment: 28	Evaluation: 4.4/5.0

Fall 2024	Engineering Mathematics II	Enrollment: 40	Evaluation: 4.6/5.0
Fall 2024	Energy System Design	Enrollment: 14	Evaluation: 5.0/5.0
Fall 2024	Capstone Design I	Enrollment: 5	Evaluation: N/A
Spr. 2024	Int. to Ref. & A/C Eng. (Team Teaching)	Enrollment: 44	Evaluation: 4.4/5.0
Spr. 2024	Basic Electric and Electronics (002)	Enrollment: 38	Evaluation: 4.5/5.0
Spr. 2024	Basic Electric and Electronics (001)	Enrollment: 51	Evaluation: 4.4/5.0
Spr. 2024	Renewable Energy	Enrollment: 39	Evaluation: 4.3/5.0

*Graduate Course

PUBLICATIONS

†Equal contribution *Corresponding author

Google Scholar: 507 citations, h-index 12 (as of Sep 2026)

Under Review

6. Dong Eun Kim, Jung Woo Shim, Sung Jong Yoo, Junmo Koo*, Joon Hyung Shim*, "Controllable Fabrication of Pt/TiO_2 Nanotube Architectures Using AAO-Templated Atomic Layer Deposition" Under review

5. Beum Geun Seo, Jae Won Lee, Junmo Koo*, "Membrane Electrode Assembly Design for Long-Endurance Underwater PEMFCs: Degradation Priorities Under Pure Oxygen and Dead-Ended Operation" Under review

4. Hyun Jin Kong, Hyung Jong Choi, Nam Il Kim, Beum Geun Seo, Doho Kim, Junmo Koo*, Kiho Bae*, Fritz B. Prinz, Joon Hyung Shim*, "Controllable Pt/C Catalyst Synthesis in Sub-Nanoscale via Liquid Sputtering for High-Performance Proton Exchange Membrane Fuel Cells" Under review

3. Myeongcho Jang, Jun Shik Kim, Seungho Yu*, Junmo Koo*, Joon Hyung Shim*, "Pourbaix-Screened Single-Cycle Atomic Layer Deposited Protective Coatings for Stabilization of Ruthenium Oxide Oxygen Evolution Catalysts" Under review

2. Ji Young Lee, Jung Woo Shim, Dong Eun Kim, Dong-hyo Kim, Dae-Hwan Lim, Young-chul Ko, Kisup Lee, Junmo Koo*, Joon Hyung Shim*, "Rapid, Energy Efficient, and Hydrogen-Free Electrochemical Oxygen Pump Achieved by Anion Exchange Membrane" Under review

1. Jayun Song†, Jung Woo Shim†, Seon Woo Park, Junmo Koo*, Joon Hyung Shim*, "Low Loading of Ir in Polymer Electrolyte Membrane Water Electrolysis Membrane Electrode Assembly Achieved by Inkjet Printing" Under review

Published

18. Jinhyuk Lim, Junmo Koo*, Joon Hyung Shim*, "Failure Modes of Polymer Electrolyte Membrane Fuel Cell Membrane-Electrode Assemblies in Air-Independent Applications" ***Case Studies in Thermal Engineering*** 76, 107455 (2025)

17. Jung Woo Shim, Junmo Koo*, Sung Jong Yoo*, Joon Hyung Shim*, "Exceptionally Stable Polymer Electrolyte Membrane Fuel Cells Enabled by One-Pot Sequential Atomic Layer Deposition of Titania and Platinum" ***ACS Sustainable Chemistry & Engineering*** 12(42), 15634-15642 (2024)

16. Jong Seon Park, Beum Geun Seo, Junmo Koo, Jin Hyuk Lim, Yong Seok Lee, Gwon Deok Han*, Fritz B. Prinz, Joon Hyung Shim*, "High-Performance Hydroxide Exchange Membrane Fuel Cell Comprising an Atomic Layer-Deposited Silver Cathode" ***Nano Letters*** 23(17), 7825-7830 (2023)

15. Beum Geun Seo, Junmo Koo, Heon Jun Jeong, Hae Wook Park, Nam Il Kim, Joon Hyung Shim*, "Performance Enhancement of Polymer Electrolyte Membrane Fuel Cells with Cerium Oxide Interlayers Prepared by Aerosol-Assisted Chemical Vapor Deposition" ***ACS Sustainable Chemistry & Engineering*** 11(29), 10776-10784 (2023) **Front Cover**

14. Junmo Koo†, Heon Jun Jeong†, Wonjoon Choi, Joon Hyung Shim*, "Localized Growth of Atomic-Layer-Deposited Platinum Catalysts on Plasma-Treated Surface of Carbon Nanotubes" ***Scripta Materialia*** 226, 115185 (2023)

13. Jong Seon Park, Beom Joon Kim, Beum Geun Seo, Gwon Deok Han, Kang-Hee Park, Junmo Koo, Hee-Deung Park*, Joon Hyung Shim*, "Hetero-Structured Palladium-Coated Zinc Oxide Photocatalysts for Sustainable Water Treatment" ***Journal of Water Process Engineering*** 45, 102488 (2022)

12. Jinhyuk Lim, Jung Woo Shim, Dong Joon Kim, Jong Seon Park, Junmo Koo, Joon Hyung Shim*, "Improvement of Fuel Cell Catalyst Performance through Zirconia Protective Layer Coating by Atomic Layer Deposition" ***Journal of Power Sources*** 498, 229923 (2021)

11. Junmo Koo†, Jun Woo Kim†, Min Sik Kim, Sungwon Yoon, Joon Hyung Shim*, "Inkjet Printing of Silica Aerogel for Fabrication of 2-D Patterned Thermal Insulation Layers" ***International Journal of Precision Engineering and Manufacturing-Green Technology*** 8(2), 445-451 (2021)

10. Jae Hyeok Lee†, Junmo Koo†, Gwon Deok Han, Jun Woo Kim, Hyung Jong Choi, Joon Hyung Shim*, "Surface Treatment of Pt Cathode Using Ceria Infiltration for High Performance Polymer Electrolyte Membrane Fuel Cells" ***International Journal of Precision Engineering and Manufacturing-Green Technology*** 8(2), 509-518 (2021)

9. Jong Seon Park, Hyung Jong Choi, Gwon Deok Han, Junmo Koo, Eun Heui Kang, Dong Hwan Kim, Kiho Bae*, Joon Hyung Shim*, "High-Performance Protonic Ceramic Fuel Cells with a $PrBa_{0.5}Sr_{0.5}Co_{1.5}Fe_{0.5}O_{5+\delta}$ Cathode with Palladium-Rich Interface Coating" ***Journal of Power Sources*** 482, 229043 (2021)

8. Junmo Koo†, Sangmin Lee†, Junmo Kim, Dong Hwan Kim, Byung-ho Choi, Taek-Soo Kim*, Joon Hyung Shim*, "Evaluating Mechanical Properties of 100nm-Thick Atomic Layer Deposited Al_2O_3 as a Free-Standing Film" ***Scripta Materialia*** 187, 256-261 (2020)

7. Dong Young Jang†, Junmo Koo†, Hyeon Rak Choi, Jun Woo Kim, Heon Jae Jeong, Fritz B. Prinz, Joon Hyung Shim*, "Coke-Free Oxidation of Methanol in Solid Oxide Fuel Cells with Heterogeneous Nickel-Palladium Catalysts Prepared by Atomic Layer Deposition" ***ACS Sustainable Chemistry & Engineering*** 8(28), 10529-10535 (2020)

6. Sangwook Park, Gwon Deok Han, Junmo Koo, Hyung Jong Choi, Joon Hyung Shim*, "Profitable Production of Stable Electrical Power Using Wind-Battery Hybrid Power Systems: A Case Study from Mt. Taegi, South Korea" ***International Journal of Precision Engineering and Manufacturing-Green Technology*** 6(5), 919-930 (2019)

5. Alireza Karimaghaloo†, Junmo Koo†, Hung-Sen Kang†, Shin Ae Song, Joon Hyung Shim*, Min Hwan Lee*, "Nanoscale Surface and Interface Engineering of Solid Oxide Fuel Cells by Atomic Layer Deposition" ***International Journal of Precision Engineering and Manufacturing-Green Technology*** 6(3), 611-628 (2019)

4. Byung Chan Yang†, Junmo Koo†, Jeong Woo Shin, Dohyun Go, Joon Hyung Shim*, Jihwan An*, "Direct Alcohol-Fueled Low-Temperature Solid Oxide Fuel Cells: A Review" ***Energy Technology*** 7(1), 5-19 (2019) **Best of Energy Technology, 2019**

3. Hyung Jong Choi, Kiho Bae, Steffen Grieshammer, Gwon Deok Han, Suk Won Park, Jun Woo Kim, Dong Young Jang, Junmo Koo, Ji-Won Son, Manfred Martin, Joon Hyung Shim*, "Surface Tuning of Solid Oxide Fuel Cell Cathode by Atomic Layer Deposition" ***Advanced Energy Materials*** 8(33), 1802506 (2018)

2. Junmo Koo, Gwon Deok Han, Hyung Jong Choi, Joon Hyung Shim*, "Wind-Speed Prediction and Analysis Based on Geological and Distance Variables Using an Artificial Neural Network: A Case Study in South Korea" ***Energy*** 93, 1296-1302 (2015)

1. You Kai Li†, Hyung Jong Choi†, Ho Keun Kim, Neoh Ke Chean, Manjin Kim, Junmo Koo, Heon Jae Jeong, Dong Young Jang, Joon Hyung Shim*, "Nanoporous Silver Cathodes Surface-Treated by Atomic Layer Deposition of Y:ZrO_2 for High-Performance Low-Temperature Solid Oxide Fuel Cells" ***Journal of Power Sources*** 295, 175-181 (2015)

RESEARCH FUNDING

Period	Title	Sponsor/ Program	Role	Amount (1,000 KRW)
Sep 2026 – Aug 2029	Direct Seawater AEMWE for Off-Grid Powering of AI Data Centers: Development of ALD-Based Pt-Free Nano 3D Electrodes	NRF/ Basic Research A	PI	120,000
Jul 2026 – Jun 2029	Basic Research Laboratory for Thermo-Fluid and Mass-Transfer Control of Direct Air Capture (DAC) in Marine Environments	NRF/ BRL	Co-I	1,500,000
Jan 2026 – Dec 2026	Particle ALD System, Analysis & Application Lab	Busan Metropolitan City/ University Open-Lab	PI	50,000
Jul 2025 – Feb 2026	Power Semiconductor Process Engineer Training Program	KMOU	PI	10,000
Jun 2025 – Nov 2025	Development of Non-Vacuum Nanoscale Thin-Film Processes for Interface/Surface Treatment	NRF/ BRIDGE3.0	PI	15,000
Sep 2024 – Dec 2024	Development of Powder Atomic Layer Deposition for High-Performance, Low-Cost Catalysts for Advanced Hydrogen Mobility	NRF/ BRIDGE3.0	PI	15,000

Period	Title	Sponsor/ Program	Role	Amount (1,000 KRW)
Aug 2024 – Nov 2024	Development of Protective Coatings via Non-Vacuum Chemical Vapor Deposition	NRF/ LINC3.0	PI	13,000
Jul 2024 – Feb 2025	Workforce Development Program for Power Semiconductor Specialized Cluster	KMOU	PI	10,000

INTERNATIONAL CONFERENCES

9. Junmo Koo "Plasma-Pretreated ALD Growth of Platinum Catalysts on Carbon Nanotubes for Polymer Electrolyte Membrane Fuel Cell Applications" ***ALD/ALE 2025***, Jeju, South Korea

8. Junmo Koo, Dong Hwan Kim, Eun Heui Kang, Joon Hyung Shim "Platinum Nanoparticles on Plasma-Pretreated Carbon Nanotubes Prepared by Atomic Layer Deposition as Polymer Electrolyte Membrane Fuel Cell Catalysts" ***236th ECS Meeting***, Atlanta, Georgia, USA, 2019

7. Junmo Koo, Dong Hwan Kim, Eun Heui Kang, Joon Hyung Shim "Atomic Layer Deposition of Platinum Nanoparticles on Plasma-Pretreated Carbon Nanotubes for Polymer Electrolyte Membrane Fuel Cell Cathode" ***The 22nd International Conference on Solid State Ionics (SSI-22)***, Pyeongchang, Korea, 2019

6. Junmo Koo, Dong Hwan Kim, Eun Heui Kang, Byung-ho Choi, Joon Hyung Shim "Atomic Layer Deposition of Platinum Nanoparticles on Plasma-Pretreated Carbon Nanotubes for High Performance Fuel Cell Catalysts" ***2019 E-MRS Spring Meeting***, Nice, France

5. Junmo Koo, Dong Young Jang, Hyeon Rak Choi, Jun Woo Kim, Joon Hyung Shim "Palladium Nanoparticles on Nickel for Direct Methanol Solid Oxide Fuel Cell Catalyst" ***AiMES 2018,*** Cancun, Mexico

4. Junmo Koo, Dong Young Jang, Hyeon Rak Choi, Jun Woo Kim, Heon Jae Jeong, Joon Hyung Shim "Atomic Layer Deposition of Palladium Nanoparticles on Nickel for Direct Methanol Solid Oxide Fuel Cell Catalyst" ***18th International Conference on Atomic Layer Deposition (ALD 2018)***, Songdo, Incheon, Korea

3. Junmo Koo, Sang Min Lee, Taek-Soo Kim, Joon Hyung Shim "Evaluating Mechanical Properties of Free-standing ALD Al_2O_3" ***17th International Conference on Atomic Layer Deposition (ALD 2017)***, Denver, Colorado, USA

2. Junmo Koo, You Kai Li, Hyung Jong Choi, Ke Chean Neoh, Dong Young Jang, Joon Hyung Shim "High-Performance Low-temperature Solid Oxide Fuel Cells by Atomic Layer Deposition of Yttria-Stabilized Zirconia on Silver Cathodes" ***PRiME 2016***, Honolulu, Hawaii, USA

1. Junmo Koo, Jae-Han Kim, Ho Keun Kim, Taek-Soo Kim, Joon Hyung Shim "Mechanical Properties of Yttria-Stabilized Zirconia Thin Films Prepared by Atomic Layer Deposition" ***15th International Conference on Atomic Layer Deposition (ALD 2015)***, Portland, Oregon, USA

INVITED TALKS

7. Junmo Koo "Atomic Layer Deposition of Nanoparticles and Thin Films for Fuel Cells" ***Mechanical Engineering Seminar, POSTECH***, Sep 2026

6. Junmo Koo "ALD-Engineered Nanoparticles and Thin Films for Fuel Cells" ***2026 Korean Society of Mechanical Engineers (KSME) Thermal Engineering Division Spring Conference***, Apr 2026

5. Junmo Koo "Hydrogen Mobility & Hydrogen Energy" ***Special Topics in Mechanical System Design, Division of Mechanical Systems Engineering, Sookmyung Women's University***, Apr 2026

4. Junmo Koo "ALD-Engineered Nanoparticles and Thin Films for Fuel Cells" ***2025 Korean Ceramic Society Fall Meeting, Y-Ceramist Symposium for Newly Appointed Researchers***, Oct 2025

3. Junmo Koo "Applications of Atomic Layer Deposition in the Energy Field" ***Korean Electrochemical Society Busan-Ulsan-Gyeongnam Branch Fall Conference***, Sep 2025

2. Junmo Koo "Semiconductor Processes & Hydrogen Energy" ***2025 Materials, Parts &***

Equipment × Future Mobility & Power Semiconductor University Technology Commercialization Showcase, May 2025

1. Junmo Koo "Semiconductor Processes & Hydrogen Energy" ***2024 1H Marine Cluster Technology Transfer Briefing***, Jun 2024

DOMESTIC CONFERENCES

3. Junmo Koo "Highly Durable Polymer Electrolyte Membrane Fuel Cell Electrodes Based on Platinum Nanoparticles by Atomic Layer Deposition" ***Korean Society for Precision Engineering (KSPE) Fall Conference***, Nov 2024

2. Junmo Koo "Atomic Layer Deposition of Pt Nanoparticles and Its Application to Polymer Electrolyte Membrane Fuel Cell Catalysts" ***Korean Institute of Surface Engineering (KISE) Fall Conference***, Nov 2024

1. Junmo Koo "Rare-Earth Oxide Thin Films Deposited by Aerosol-Assisted Chemical Vapor Deposition" ***Korean Society of Marine Engineering (KOSME) Fall Conference***, Oct 2024

ISSUED PATENTS

6. Joon Hyung Shim, Jong Seon Park, Beum Geun Seo, Junmo Koo, Atomic layer deposition apparatus, 10-2691300, South Korea

5. Joon Hyung Shim, Jong Seon Park, Heon Jun Jeong, Junmo Koo, Atomic layer deposition system and electrochemical measuring method using the same, 10-2359510, South Korea

4. Joon Hyung Shim, Jong Seon Park, Junmo Koo, Keun Hee Kim, System for flow retention of atomic layer deposition processing using retention valve, 10-2361065, South Korea

3. Joon Hyung Shim, Hee-Deung Park, Jong Seon Park, Beom Joon Kim, Kang Hee Park, Junmo Koo, Heterogeneous catalyst structure and method for manufacturing the same,

10-2384204, South Korea

2. Joon Hyung Shim, Gwon Deok Han, Ke Chean Neoh, Junmo Koo, Water-based ceramic ink containing binder for enhanced spatial resolution, 10-2367507, South Korea

1. Joon Hyung Shim, Gwon Deok Han, Ke Chean Neoh, Junmo Koo, Water-based ceramic ink for inkjet printing, 10-2367506, South Korea

PATENT APPLICATIONS

5. Sanghyun Lee, Junmo Koo, Shear-thickening fluid composition comprising silica nanoparticles surface-modified by atomic layer deposition and method for preparing the same, 10-2026-0055417, South Korea

4. Junmo Koo, Sanghyun Lee, Particle-aligning microfluidic chip for atomic layer deposition, 10-2025-0152000, South Korea

3. Junmo Koo, Deposition apparatus for powder substrates for atomic layer deposition, 10-2025-0024986, South Korea

2. Junmo Koo, Yttrium oxide thin film and method for manufacturing the same, 10-2024-0174323, South Korea

1. Joon Hyung Shim, Jong Seon Park, Junmo Koo, Plasma supply device for atomic layer deposition, plasma supply system, atomic layer deposition apparatus, and control method thereof, 10-2020-0019808, South Korea

TECHNICAL SKILLS

Thin film fabrication techniques: atomic layer deposition (ALD), chemical vapor deposition (CVD), inkjet printing and sputtering

Thin film characterization techniques: transmission electron microscopy (TEM), electron energy loss spectroscopy (EELS), atomic force microscopy (AFM), X-ray reflectometry (XRR), X-ray photoelectron spectroscopy (XPS), X-ray diffractometer (XRD) and scanning electron microscopy (SEM)

Vacuum system design: ALD chambers and gas-environment chambers

JOURNAL REVIEWER

Journal of Mechanical Science and Technology (Springer Nature) 2024, 2025

JMST Advances (Springer Nature) 2025

PROFESSIONAL SERVICE

Planning & Research Committee Member

Civil-Military Technology Cooperation Program 2026

Technical Advisory/Review Committee Member

Chungbuk Development Corporation Mar 2025 – Mar 2027

Evaluation Committee Member

Eco-Friendly Ship Certification Program Jan 2025 – Dec 2025

Building Committee Member

Gangseo-gu, Busan Metropolitan City Nov 2024 – Nov 2025

Expert Advisory Group Member

Gangwon State Oct 2024 – Oct 2026

Technical Advisory Committee Member

Changwon City Sep 2024 – Oct 2026

Innovative Product Evaluator, SME Technology Market

Korea Expressway Corporation Sep 2024 – Dec 2025

11th Technical Advisory Committee Member

Busan Metropolitan Corporation Sep 2024 – Aug 2026

Evaluator, Technology Doctor Program

Gyeonggi Province Jun 2024 – Jun 2026

11th Technical Advisory Committee Member

Gimhae City Apr 2024 – Mar 2026

Review Committee Member

Eco-Friendly Certified Ship Deployment Support Program Apr 2024 – Feb 2025

CURRENT STUDENTS

Dong Ah Lee | Undergraduate

THESIS COMMITTEES SERVED

Seunghun Lee | Seong Min Lee (MS at Korea Maritime & Ocean University) Feb. 2025

Han-sik Chae | Bong-Hee Lee (PhD at Korea Maritime & Ocean University) Aug. 2024

REFERENCES

Prof. Joon Hyung Shim (PhD advisor) School of Mechanical Engineering, Korea University shimm@korea.ac.kr

Dr. Jong-Ho Lee (Post-doc PI) Center for Energy Materials Research, Korea Institute of Science and Technology jongho@kist.re.kr

Prof. Jihwan An (Thesis committee) Department of Mechanical Engineering, POSTECH jihwanan@postech.ac.kr

Prof. Youngseung Na (Thesis committee) Department of Mechanical and Information Engineering, University of Seoul ysna@uos.ac.kr